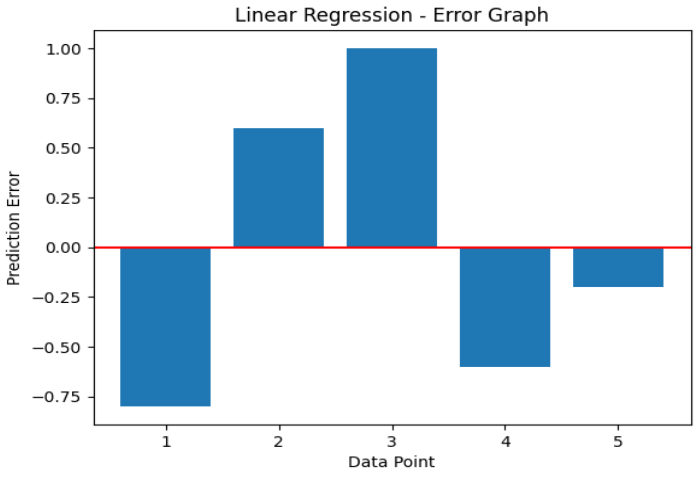
EXP 8: Linear Regression

CODE

```
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
X = [[1], [2], [3], [4], [5]]
y = [2, 4, 5, 4, 5]
model = LinearRegression()
model.fit(X, y)
pred = model.predict(X)
errors = [a-b for a, b in zip(y, pred)]
plt.bar([1, 2, 3, 4, 5], errors)
plt.axhline(0, color="red")
plt.xlabel("Data Point")
plt.ylabel("Prediction Error")
plt.title("Linear Regression - Error Graph")
plt.show()
print("Prediction for X=6:", round(model.predict([[6]])[0], 2))
```

OUTPUT



Prediction for X=6: 5.8